Shemonto Das

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Highlights of Qualification

AI Engineer with three years of experience delivering end-to-end machine learning and AI solutions that accelerate product development and business growth for early-stage startups to established companies. Adept at building LLM-powered client applications, integrating retrieval-augmented generation (RAG) pipelines, and deploying agentic applications that improve customer engagement and operational efficiency. Skilled in data science, analytics, predictive modeling, and machine learning model development, with a strong focus on translating complex data challenges into business value. Recognized for academic and industry contributions through multiple publications, awards, and leadership roles.

Education

MSc. in Computer Science, Memorial University of Newfoundland, St.John's, NL — May 2024

BSc. in Computer Science & Engineering, BRAC University, Dhaka, Bangladesh — January 2021

Skills

- **Language and Frameworks**: Python, Java, HTML, Keras, Pandas, Scikit-learn, Streamlit, LangChain, LangGraph, FastAPI, Docker, MS Office.
- **Soft skills**: Scientific Research, Academic Writing, Presentation, Critical thinker, Team player, Leadership.

Experience

AI Engineer, **Genesis AI Garage** – St. John's, NL, Canada — February 2025 – Present

- Spearheaded AI initiatives across **5+ early-stage startups** and **established companies**, delivering scalable AI solutions that accelerated **time-to-market** and enhanced business outcomes for diverse clients.
- Designed and experimented **predictive models** for **classification, forecasting, and anomaly detection**, incorporating automated hyperparameter tuning for optimal performance.
- Developed and deployed **LLM-powered client applications with RAG and agentic frameworks (LangChain, LangGraph)**, driving improvements in **customer engagement KPIs** and **reducing manual operational workload**.
- Delivered **secure, cloud-based APIs and LLM services**, enabling seamless integration into client systems **with clear technical documentation** to ensure self-sufficiency post-deployment.
- Prepare and visualize **structured/unstructured data** to derive actionable feature insights for clients.
- Collaborated directly with founders, product teams, and business leaders, translating **high-level business goals into practical AI solutions** that unlocked new revenue opportunities.
- Consistently adopted and customized emerging AI tools (OpenAI, Gemini, Ollama) to unlock **new revenue channels** and product features for clients.

Data Quality Specialist, **SiftMed** – St. John's, NL, Canada — October 2024 – Present

- Streamlined digital document organization, labeling accuracy and proactively resolved data quality issues.
- Collaborated with cross-functional teams **using Agile methodologies** to **implement data management strategies, enhancing efficiency and data governance**

Data Scientist, **Instrumar Limited** – St. John's, NL, Canada — June 2024 – September 2024

- Analyzed and explored **time series patterns** to develop, optimize, and interpret AI models that improved **fault detection accuracy** in manufacturing processes.
- Developed and automated scalable pipeline for data querying, cleaning, feature engineering and visualization.
- Reduced **data annotation cost and time** for Instrumar by **40%**, making **fault classification models** robust and efficient.

Data Science Intern, **Instrumar Limited** – St. John's, NL, Canada May 2022 – April 2024

- Interpreted complex manufacturing data into actional insights, **improving data-driven decision-making**.
- Optimized **data preprocessing and feature engineering** to enhance anomaly detection and classification models.

Projects

LLM-Powered Chatbots with RAG and Agentic Frameworks (Python, LangGraph, LangChain, Gemini, FastAPI, Docker, GCP)

- Built **multi-agent conversational AI systems** using LangGraph for dynamic workflow orchestration.
- **Automated pipelines for RAG**, including data preprocessing, vector embedding & retrieval on structured/unstructured data.
- Deployed the entire solution on **Google Cloud** using **FastAPI** and **Dockerized microservices**, ensuring scalability, high availability, and **seamless API integrations**.
- Optimized performance through streamlined data pipelines, response caching, and asynchronous processing, **reducing API latency and operational costs**.

Predicting Cross-sell Insurance: Explainable AI Project (Python, Scikit-learn, Streamlit) **Link**

- Developed a cross-selling model for health & vehicle insurance with **0.98 F1 score**, 2nd place in competition.
- Developed a Streamlit dashboard to visualize model predictions and **integrate LIME and SHAP** for reporting enhanced interpretability, **ensuring bias, fairness assessments and ethical AI**.

Loan Approval Prediction | Kaggle Playground Series 2024 (Python, Pandas, Matplotlib, Scikit-learn) **Link**

- Developed loan approval prediction model using CatBoost with extensive feature engineering & data balancing.
- Achieved **0.96 AUC on Kaggle**; optimized model via hyperparameter tuning, learning curve & ROC analysis.

Instrumar's Machine Learning Pipeline for Fault Classification (Python, Pycaret, Scikit-learn, Graphviz) **Link**

- Reduced **quality downgrades by 30%** through automated fault classification, improving detection efficiency.
- Developed a **novel frequency-based algorithm to tackle data imbalance**, boosting accuracy for Instrumar Limited.
- Ensembled and stacked ML models to enhance performance and predictive accuracy.

Active Learning Framework for Texture Classification (Python, Scikit-learn, Pandas, Scipy) **Link**

- Developed framework with Uncertainty Sampling & Query by Committee strategies for optimized classification.
- Validated machine learning model results for texture classification using **Wilcoxon statistical hypothesis testing**.

Publications

- Active Learning Strategies for Robotic Tactile Texture Recognition Tasks, Journal: Frontiers in Robotics and AI, section Robot Learning and Evolution.
- Unbalanced Fault Classification Using Active Learning in Synthetic Fiber Manufacturing Process, IEEE Systems Conference 2024.

Awards

- Awarded "**Fellow of the School of Graduate Studies**" title by Memorial University, 2024.
- Graduated with the **Highest Distinction** from Memorial University of Newfoundland.
- **Graduate Research Award 2023** for strong research contribution, Computer Science, Memorial University.
- Awarded for **Best Graduate Talk** at the SEA 2023 conference by the Faculty of Science, Memorial University.
- Presented accepted research paper at IEEE Systems Conference 2024, Montreal, Canada.